KLAi

Derivatives: the slope of the loss tells a model which way to change a number

A model has a number it can adjust. How does it know whether to raise it or lower it? (Session 10 answers how much.)

Mathematics for AI and Machine Learning | Session 7 of 18 | Calculus and optimisation

KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Find the slope of a curve at a point as change in output divided by change in input, and check it with a finite difference

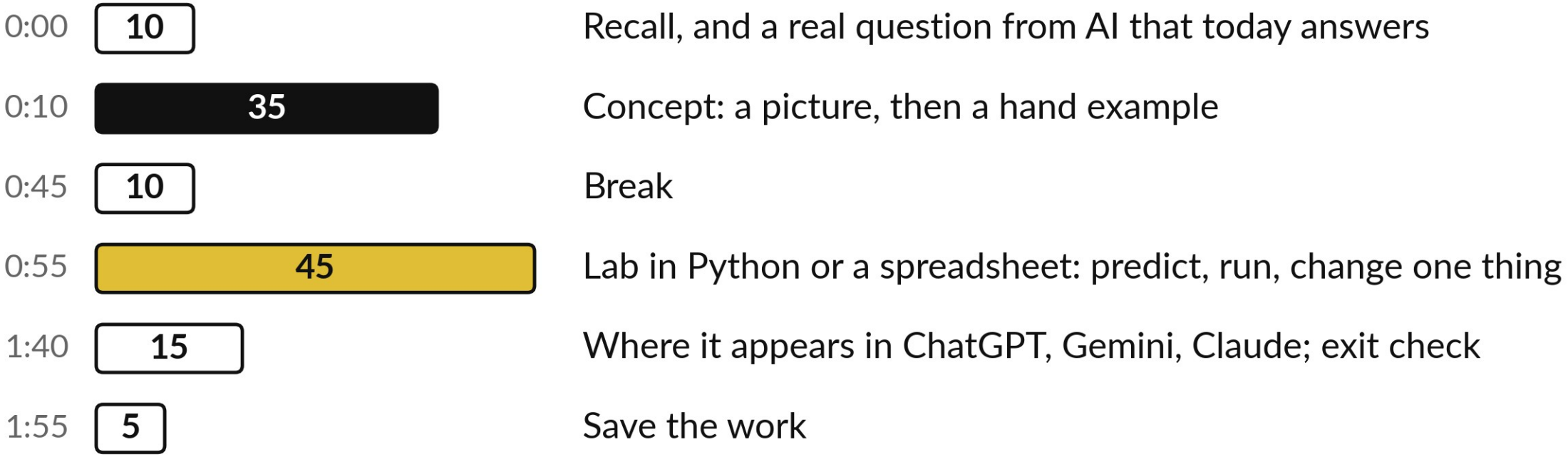


Use the power, sum, product, quotient, exponential and logarithm rules



Take the derivative of a squared-error loss for a one-parameter model and say which way to move the parameter

Your two hours: slope, rules, the loss, then the lab



Pictures and hand examples first, then the lab on the same numbers in Python or a spreadsheet.

Three questions from last time

1. What three pieces does the singular value decomposition (SVD) split a matrix into?

2. What does the first principal component in PCA point along?

3. How can keeping only the largest singular values compress a matrix?



Answer aloud first. Then we check together.

Answers to the three questions from Session 6

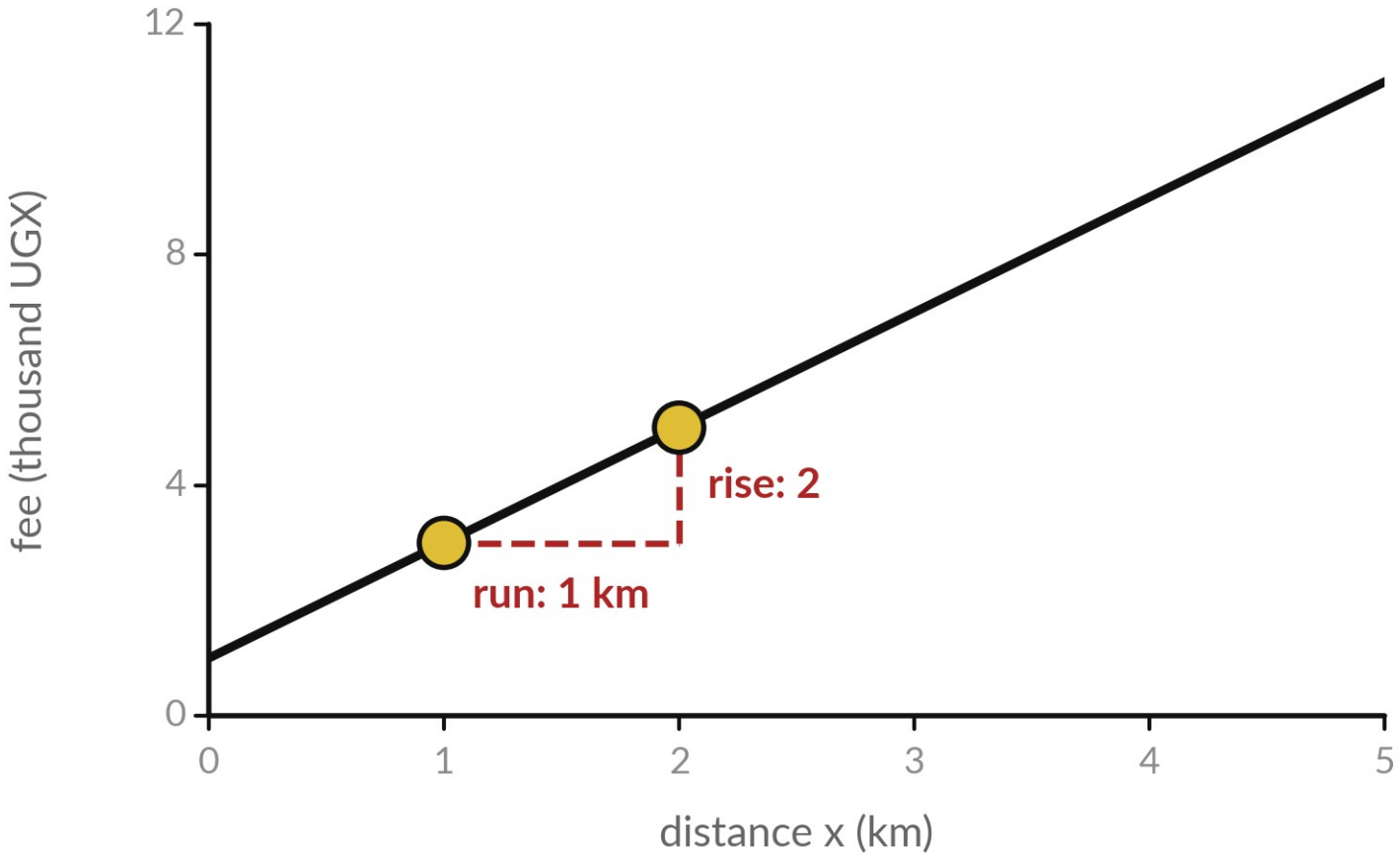
Question	Short answer
What three pieces does the singular value decomposition (SVD) split a matrix into?	U, a diagonal matrix of singular values called Sigma, and V transpose. Read right to left: rotate, stretch along the axes, rotate again.
What does the first principal component in PCA point along?	The direction in which the data vary the most. Each later component is at right angles to the ones before and captures less variation.
How can keeping only the largest singular values compress a matrix?	Keep the k largest and drop the rest. The result is a low-rank approximation that stores far fewer numbers. LoRA fine-tuning trains small matrices in the same spirit.

If one answer surprised you, look back at Session 6 before the lab. Session 9 uses matrices again.

Words we use today

	Word	What it means
	**Derivative**	The slope of a function at one point: how fast the output changes when the input changes. Written f-prime of x or df/dx.
	**Tangent line**	The straight line that just touches a curve at one point. Its slope is the derivative there.
	**Loss function**	One number that says how wrong a model is on the data. Smaller is better. Here: the mean squared error.
	**Finite difference**	An estimate of a slope from two nearby points: change in output divided by a small change in input.
	**Minimum**	A lowest point of a curve. At a minimum, or a maximum, the slope is zero.

The slope of a straight line is the change in output divided by the change in input



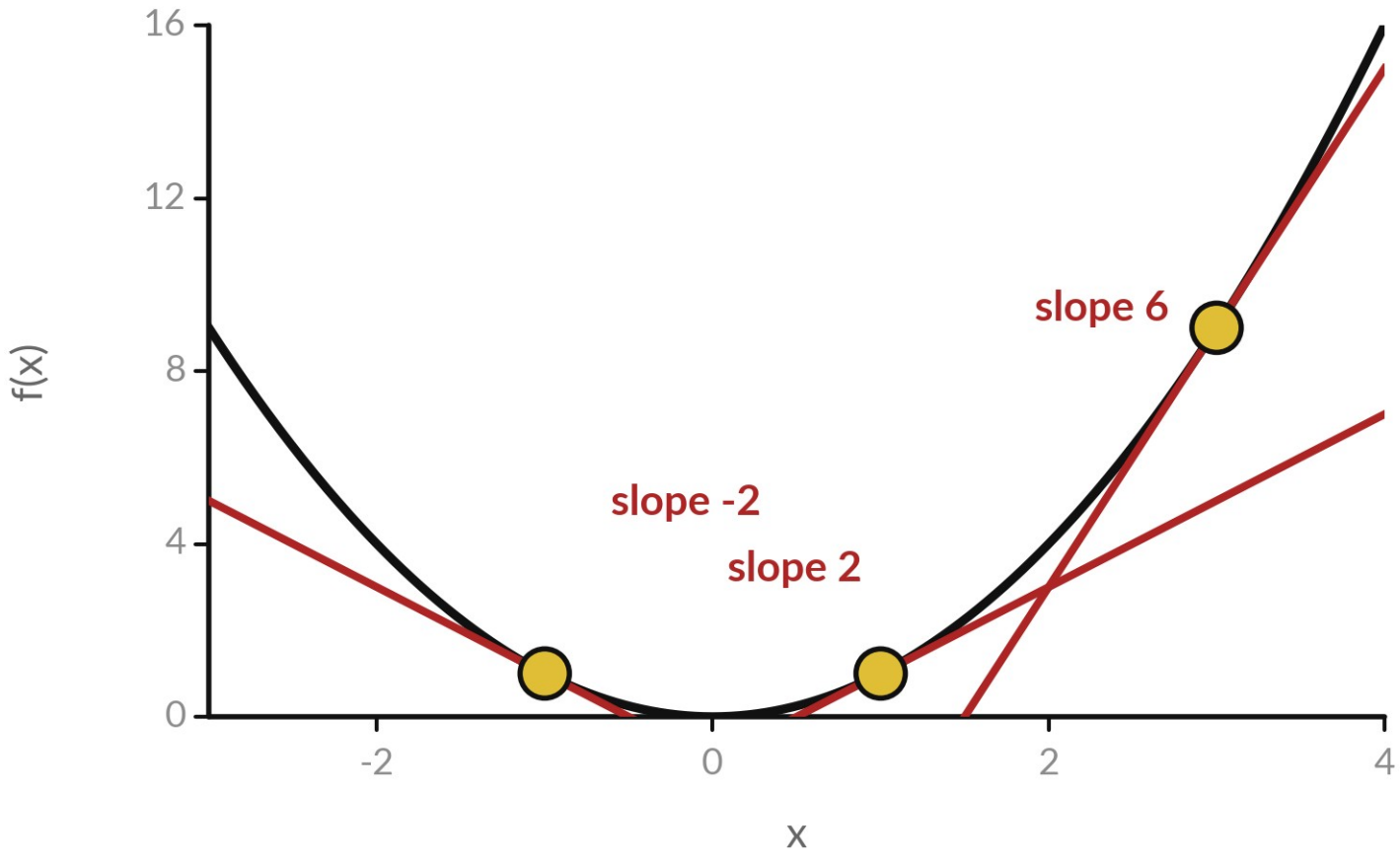
An illustrative delivery fee rises by 2 thousand UGX for every extra kilometre.

The red dashes are the run (change in input) and the rise (change in output).

Slope = rise divided by run = 2 divided by 1 = 2.

On a straight line the slope is the same everywhere.

A curve has a different slope at every point, and a tangent line shows it



The curve is f(x) = x squared. Each of the three red lines is a tangent line: the straight line that just touches the curve at one point.

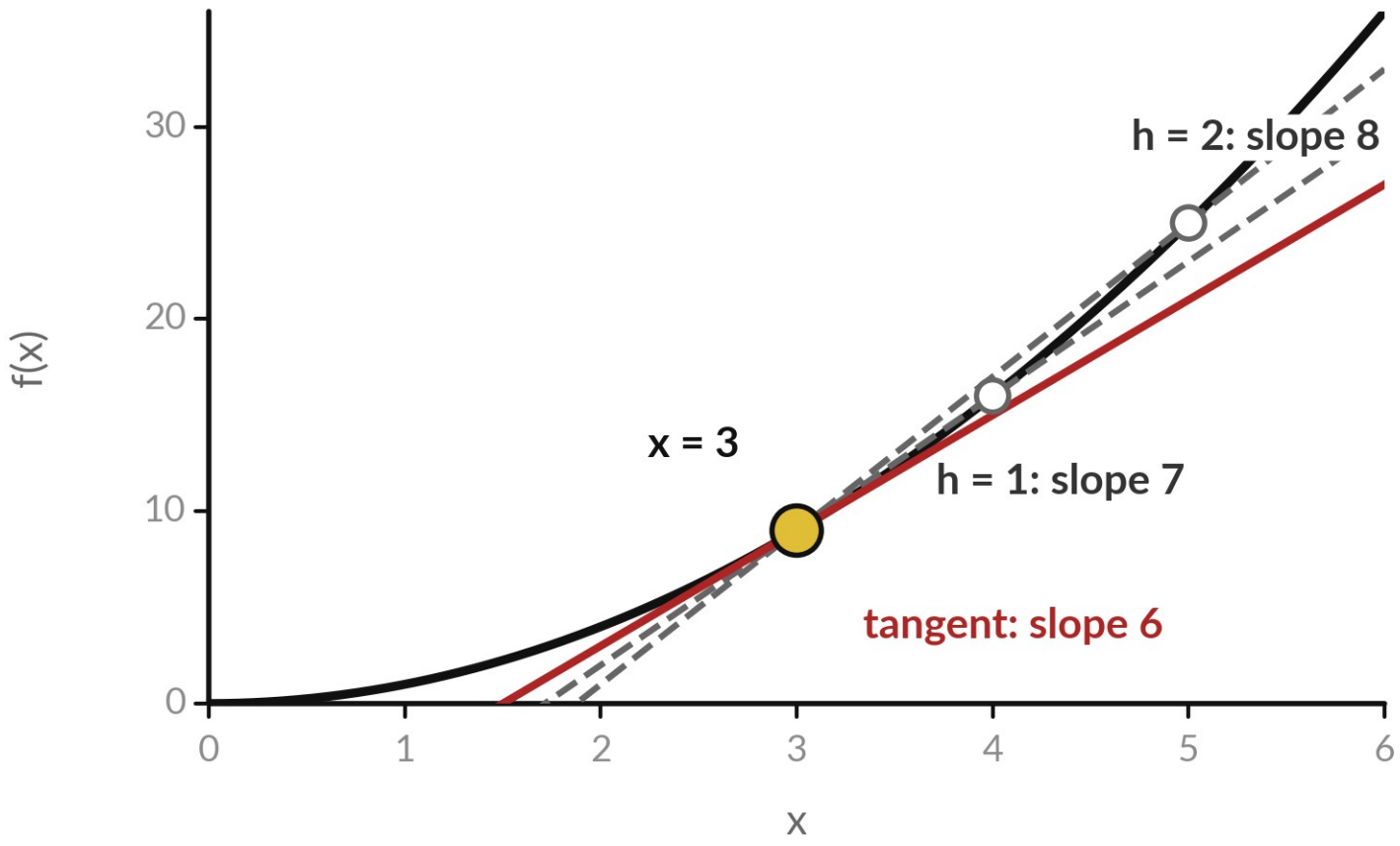
At x = -1 the slope is -2: the curve is falling.

At x = 1 the slope is 2.

At x = 3 the slope is 6: the curve is climbing steeply.

The slope of the tangent line is called the derivative.

The limit idea in one picture: shrink the step h and the secant line becomes the tangent line



A secant line joins the points at x = 3 and x = 3 + h. Its slope is (f(3 + h) minus f(3)) divided by h.

Make h smaller and the second point slides towards the first. The secant turns into the tangent.

The value the slopes settle on is the derivative. Here it is 6. This is the limit idea.

Worked by hand: for f(x) = x squared at x = 3, the slopes settle on 6

Step h	f(3 + h)	f(3)	Slope = (f(3 + h) minus f(3)) divided by h
2	25	9	8
1	16	9	7
0.5	12.25	9	6.5
0.1	9.61	9	6.1
0.01	9.0601	9	6.01
0.001	9.006	9	6.001

Each slope equals 6 + h, so it approaches 6 as h shrinks. In the lab you build this table yourself, in Python and in a spreadsheet.

The derivative is the limit of the secant slope as h goes to zero

Definition

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Other notation

$$df/dx = f'(x)$$

Symbol	What it means	In our data
f(x)	The function whose slope we want. It turns an input x into an output.	x squared
h	A small step in the input. The limit asks what happens as h goes to 0.	2, 1, 0.5, 0.1 ...
f'(x) or df/dx	The derivative: the slope of the tangent line at x. Read df/dx as change in f per change in x.	f'(3) = 6

Both notations mean the same. Papers and library manuals use df/dx.

Three rules cover polynomials: the power rule, the constant multiple rule and the sum rule

Power rule

$$\frac{d}{dx} x^n = n\, x^{n-1}$$

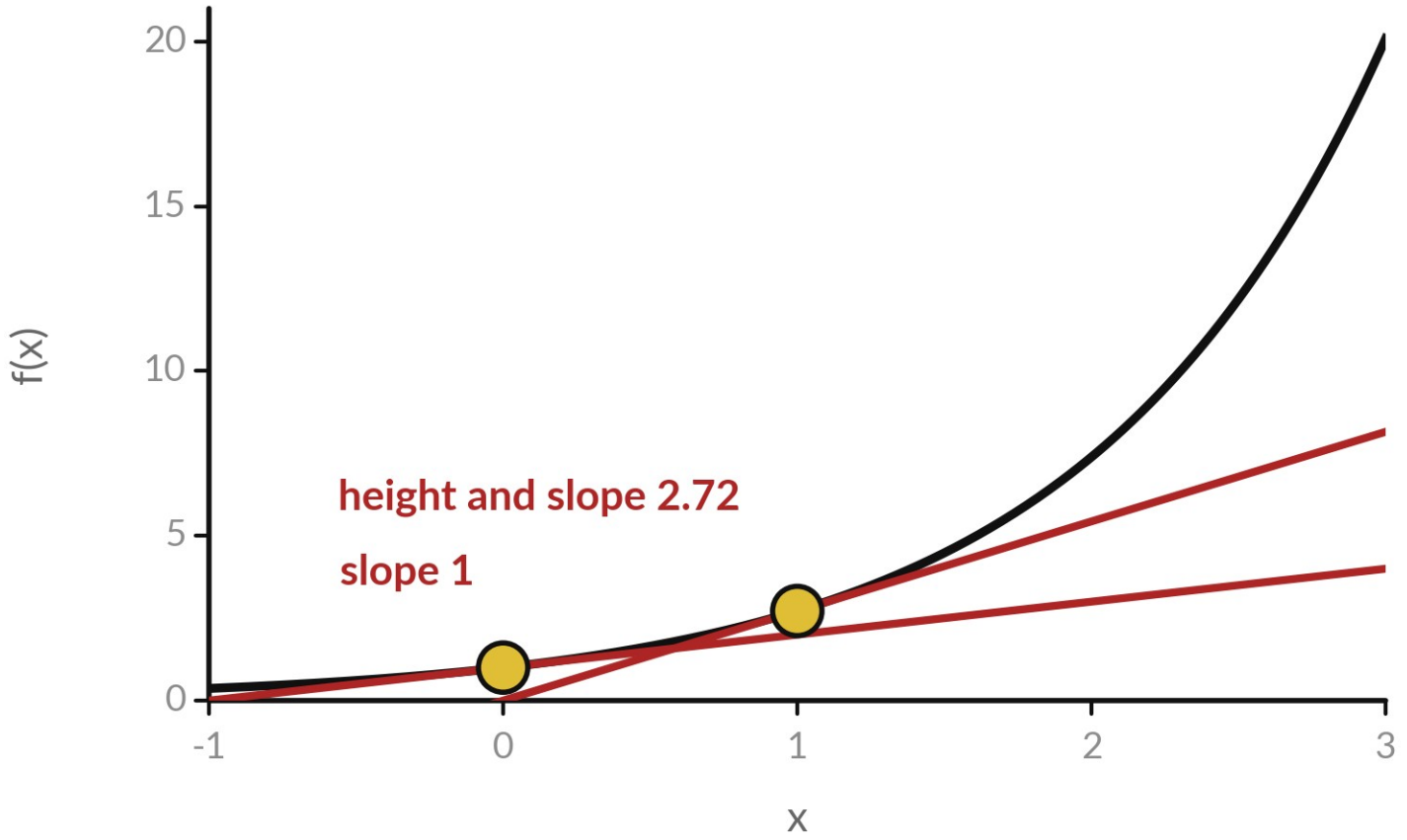
Constant multiple and sum

$$(c\, f + g)' = c\, f' + g'$$

Symbol	What it means	In our data
n	The power. Bring n down in front and lower the power by one.	x cubed: slope 3 x squared
c	A fixed number multiplying a function. It stays in front.	the 3 in 3 x squared
f, g	Any two functions. The slope of a sum is the sum of the slopes.	f = 3 x squared, g = 2x

Example: f(x) = 3 x squared + 2x has f-prime(x) = 6x + 2. At x = 2 the slope is 14.

The exponential e to the x is its own slope: at every point the slope equals the height

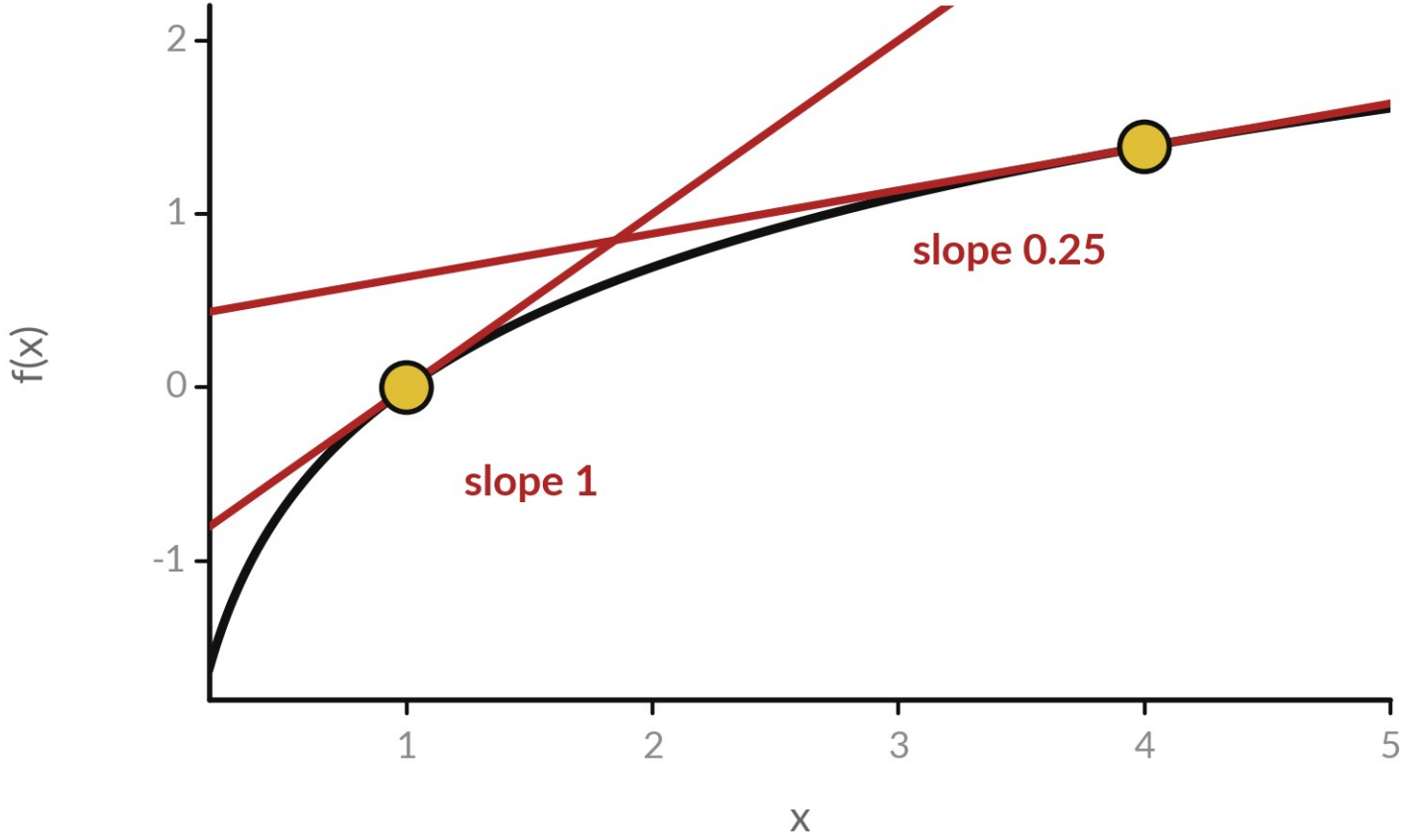


e is about 2.718. The curve y = e to the x is the one whose slope at each point equals its height there.

Rule: the derivative of e to the x is e to the x.

With a constant inside, e to the 2x has slope 2 times e to the 2x. Session 9 shows why.

The natural logarithm flattens out: its slope at x is 1 divided by x

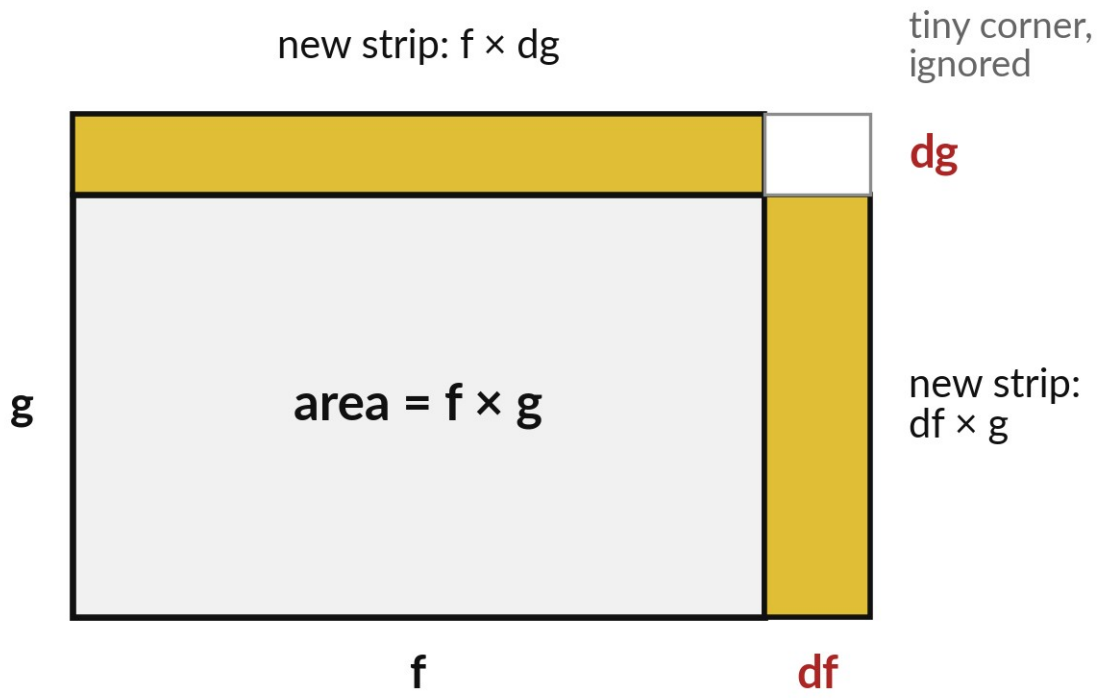


ln x is the natural logarithm from Session 1. It is the inverse of e to the x.

Rule: the derivative of ln x is 1 divided by x. At x = 1 the slope is 1. At x = 4 it is only 0.25.

Cross-entropy (Session 17), the loss that language models are trained with, is built from logarithms.

The product rule: when two quantities both change, the area of the rectangle grows in two strips



A rectangle has width f and height g. If f grows by df and g grows by dg, the area gains two strips.

Product rule: the derivative of f times g is f-prime times g, plus f times g-prime.

The tiny corner, df times dg, shrinks to nothing in the limit.

Product and quotient rules, worked with numbers at x = 1

Product rule

$$(fg)' = f'g + fg'$$

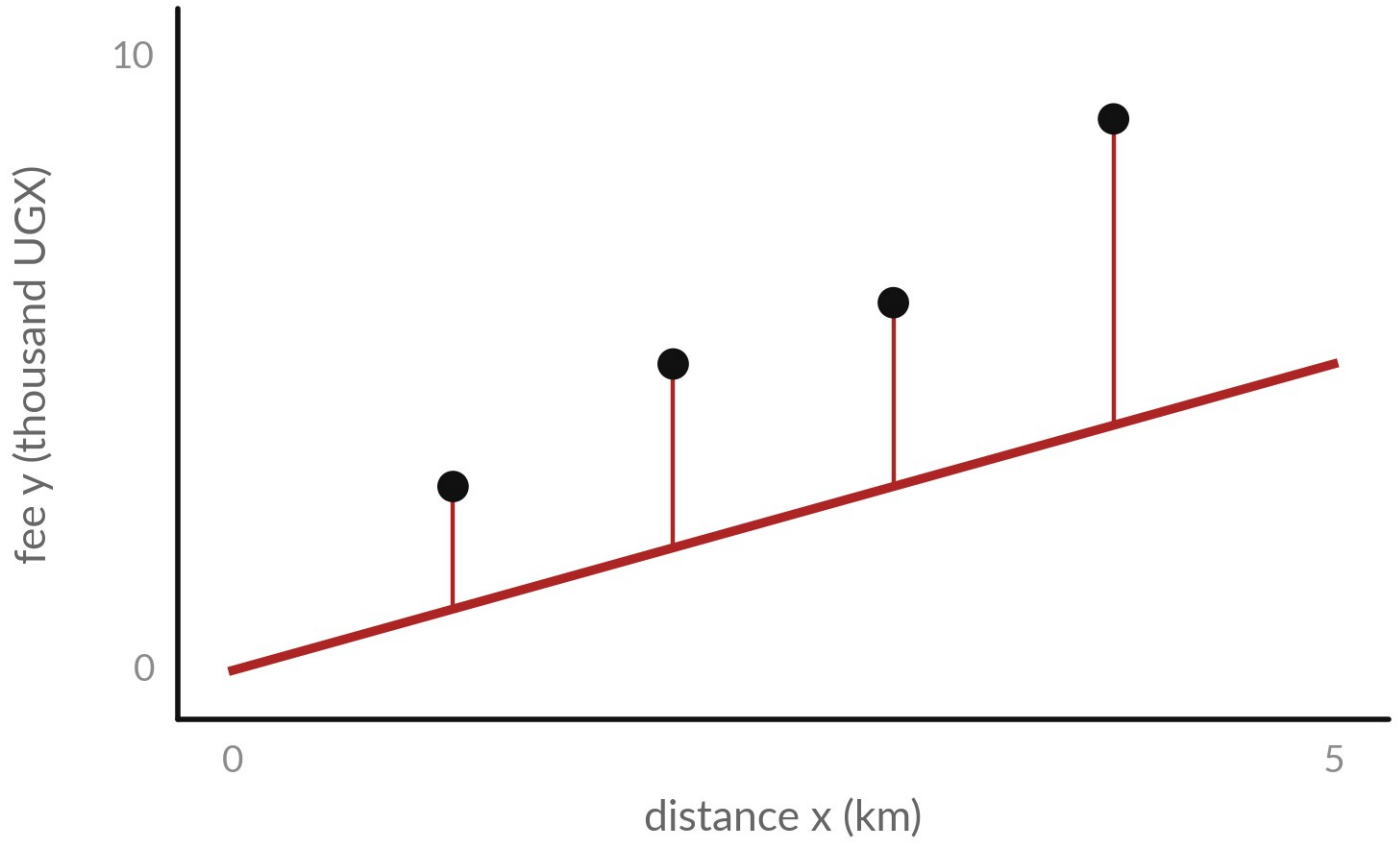
Quotient rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Symbol	What it means	In our data
f, g	Two functions of x. In the quotient rule g must not be zero.	f = x squared
f', g'	Their derivatives, each found with the rules before.	f' = 2x, g' = e to the x
Product at x = 1	x squared times e to the x: (2x + x squared) e to the x.	3e = 8.155
Quotient at x = 1	x squared divided by (x + 1): (2x(x + 1) minus x squared) divided by (x + 1) squared.	(4 minus 1) / 4 = 0.75

The quotient rule follows from the product and chain rules (Session 9). Use it as given and check it in the lab.

A loss function turns the errors of a model into one number



Each dot is one delivery (illustrative, made-up numbers). The model is y-hat = w x, with one adjustable number, w. Here w = 1.

The red segments are the errors: the gap between each dot and the line.

The loss function squares each error and takes the mean. With w = 1 the loss is 11.75. With w = 2 it is only 0.75.

The squared-error loss for a one-parameter model, with real numbers

Loss (mean squared error)

$$L(w) = \frac{1}{n} \sum_{i=1}^{n} (w\,x_i - y_i)^2$$

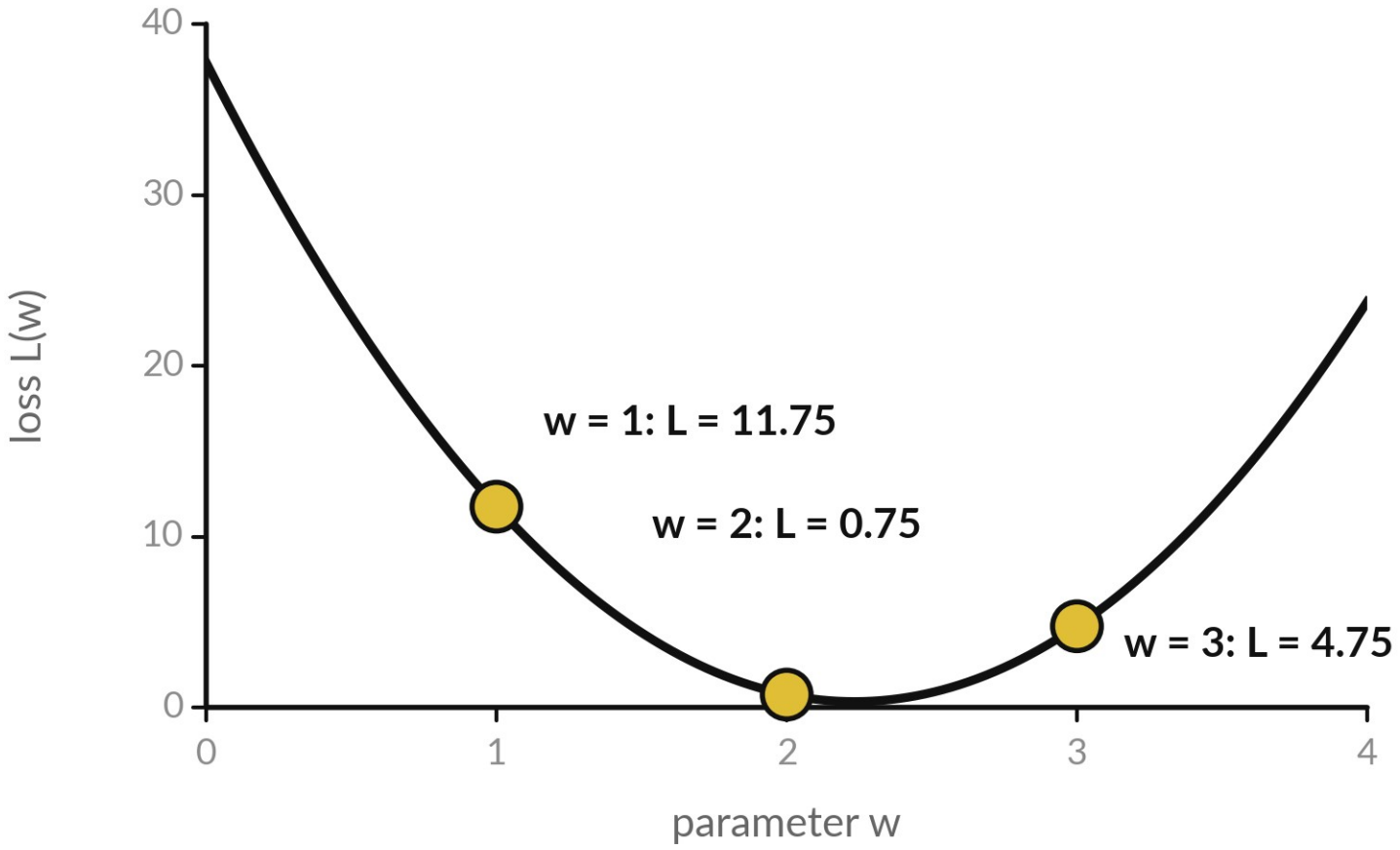
Our data at w = 1

$$L(1) = (4 + 9 + 9 + 25)/4 = 11.75$$

Symbol	What it means	In our data
L(w)	The loss: the mean of the squared errors, for a given w.	L(1) = 11.75, L(2) = 0.75
w	The one parameter (adjustable number) of the model y-hat = w x.	fee per kilometre
x_i, y_i	Distance and fee of delivery i. Subscript i counts the rows.	x = 1, 2, 3, 4; y = 3, 5, 6, 9
n	The number of rows.	4 deliveries

The errors at w = 1 are -2, -3, -3 and -5. Squared: 4, 9, 9, 25.

The loss is a curve in w, and its lowest point is the best parameter

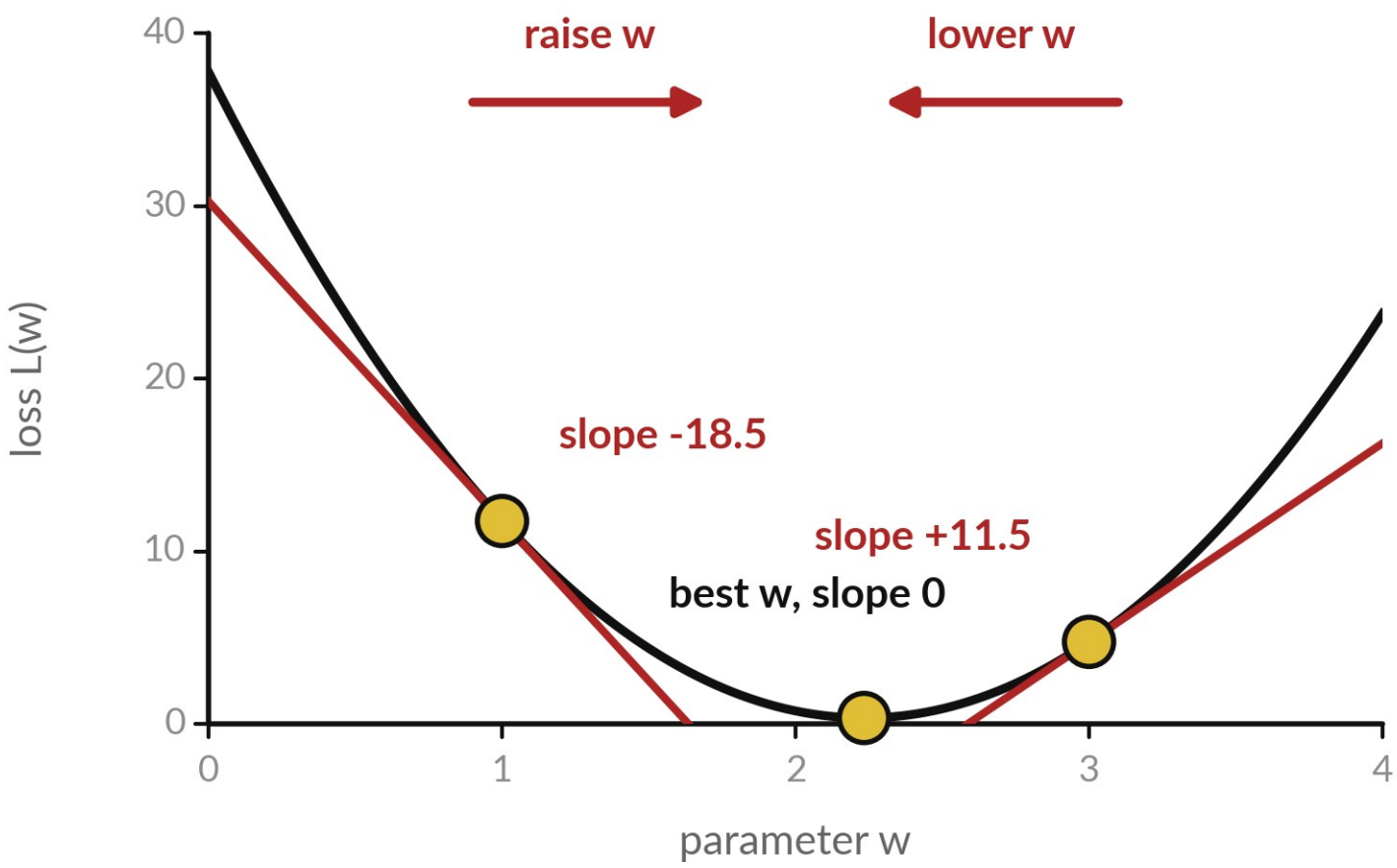


Try many values of w and plot the loss L(w). The curve is a bowl.

Each gold dot is one value of w and its loss.

Training a model means finding the w at the bottom of the bowl.

The slope of the loss tells the model which way to change w



At w = 1 the slope of the loss is -18.5: negative, so the loss falls as w grows. At w = 3 it is +11.5: positive, so the loss falls as w shrinks.

Move w opposite to the sign of the slope. The third gold dot, w = 2.233, is the best w, where the slope is 0.

A steeper slope means the loss changes more for the same step.

The derivative of the squared-error loss, and its value at w = 1

Derivative of the loss

$$\frac{dL}{dw} = \frac{2}{n} \sum_{i=1}^{n} x_i (w\, x_i - y_i)$$

Our data at w = 1

$$dL/dw = (2/4) \times (-37) = -18.5$$

Symbol	What it means	In our data
dL/dw	The slope of the loss curve: how fast the loss changes with w.	-18.5 at w = 1; 11.5 at w = 3
2 / n	The 2 comes from the power rule on the square. n is the number of rows.	2 / 4
x_i (w x_i - y_i)	Each error times its input x_i. The factor x_i appears because w is multiplied by x_i. Session 9 explains this with the chain rule.	products add up to -37

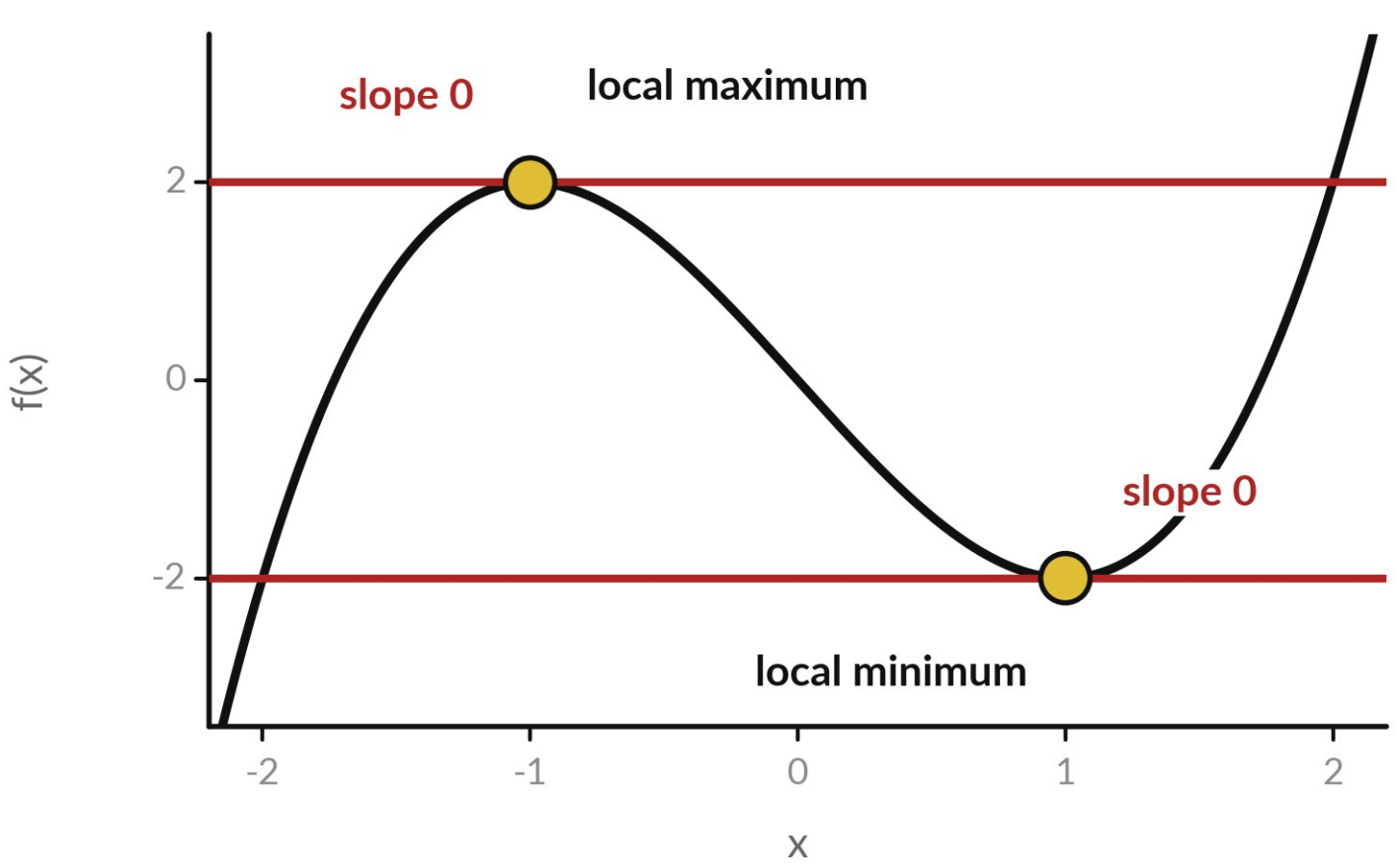
In the lab you check this formula against a finite difference.

Worked by hand at w = 1: the sum of x times error is -37, so the slope is -18.5

Row i	Distance x	Fee y	Prediction w x	Error (prediction minus y)	x times error
1	1	3	1	-2	-2
2	2	5	2	-3	-6
3	3	6	3	-3	-9
4	4	9	4	-5	-20
Sum					-37

Slope = 2/4 times (-37) = -18.5. Check by finite difference: L(1.001) = 11.7315, so (L(1.001) minus L(1)) divided by 0.001 = -18.49, close to -18.5.

At a minimum or a maximum the slope is zero



For f(x) = x cubed minus 3x the slope is 3 x squared minus 3. It is zero at x = -1 and at x = 1.

At x = -1 the curve tops out, a local maximum. The slope goes from positive to negative.

At x = 1 it bottoms out, a local minimum. The slope goes from negative to positive.

Setting the slope of the loss to zero gives the best w directly

Slope equals zero

$$\frac{dL}{dw} = 0 \Rightarrow w^* = \frac{\sum x_i y_i}{\sum x_i^2}$$

Our data

$$w^* = 67/30 = 2.233$$

Symbol	What it means	In our data
w*	The best w: the value where the loss is lowest. Read it as w-star.	2.233
sum of x_i y_i	Add the product of each distance and fee.	3 + 10 + 18 + 36 = 67
sum of x_i squared	Add the square of each distance.	1 + 4 + 9 + 16 = 30
L(w*)	The loss at the best w.	0.342

This is the Session 4 least-squares answer for a line through the origin.

Two mistakes to avoid when you read a derivative

Reading the sign backwards

Negative slope: raise w.

Positive slope: lower w.

Move opposite to the slope.

Assuming a minimum

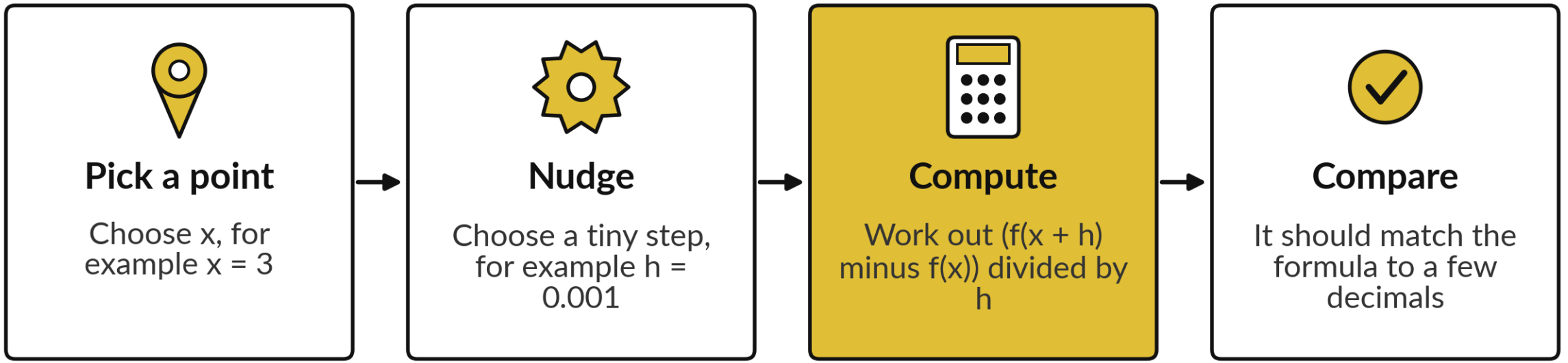
It can be a maximum or a flat spot.

Look at the loss on both sides.

Our bowl has one bottom.

Check the direction and the type of point.

Check a derivative with numbers: nudge the input and divide the change in output by the nudge



This is a finite difference: an estimate of a slope from two nearby points. Professionals check every hand-derived derivative this way.

Forward and central differences, with numbers

Forward difference

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

Central difference

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

Symbol	What it means	In our data
h	The step. About 0.001 to 0.00001 (1e-3 to 1e-5) works well. Much smaller values give rounding noise.	h = 0.001
x squared at x = 3	Exact slope 6.	forward 6.001, central 6.000
e to the x at x = 1	Exact slope 2.71828.	forward 2.71964, central 2.71828

The central difference looks on both sides of x. Use it for checks.

Read the recipe before you write it: a function that estimates a slope

```
def slope(f, x, h=1e-5):
    return (f(x + h) - f(x - h)) / (2 * h)

square = lambda x: x ** 2
print(round(slope(square, 3), 4))
```

```
6.0
```

In plain words

Line 1: slope takes a function f, a point x and a small step h.

Line 2: the central difference formula from the last slide.

Line 4: lambda is a short way to write f(x) = x squared.

Line 5: slope at x = 3, rounded to 4 decimals. It prints 6.0.

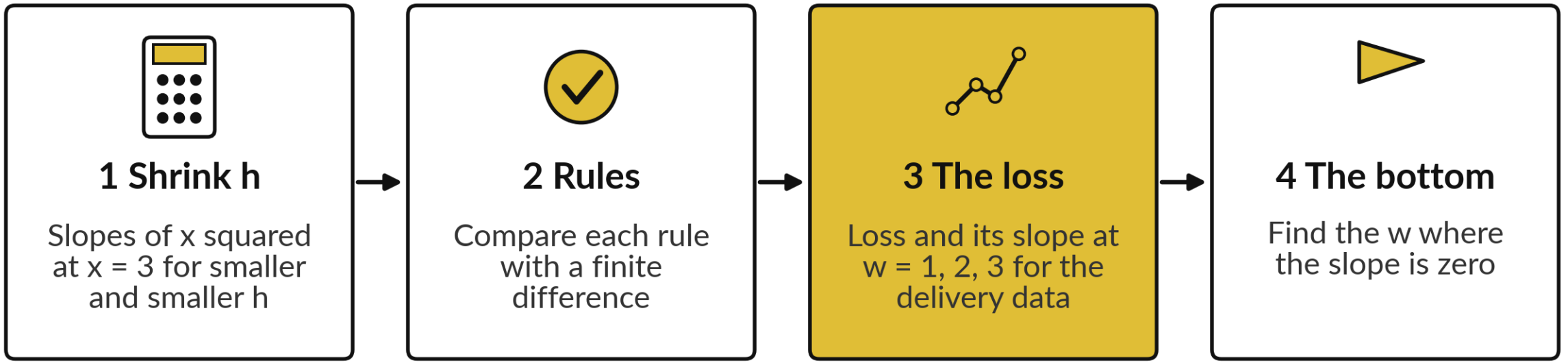
The same function checks any derivative in this course.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has four stages, in Python and in the spreadsheet



Python: Steps 1 to 7 of the notebook. Spreadsheet: sheets Shrink, Rules and Loss. In every step predict first, then run, then change one thing.

Check your numbers against these before you move on

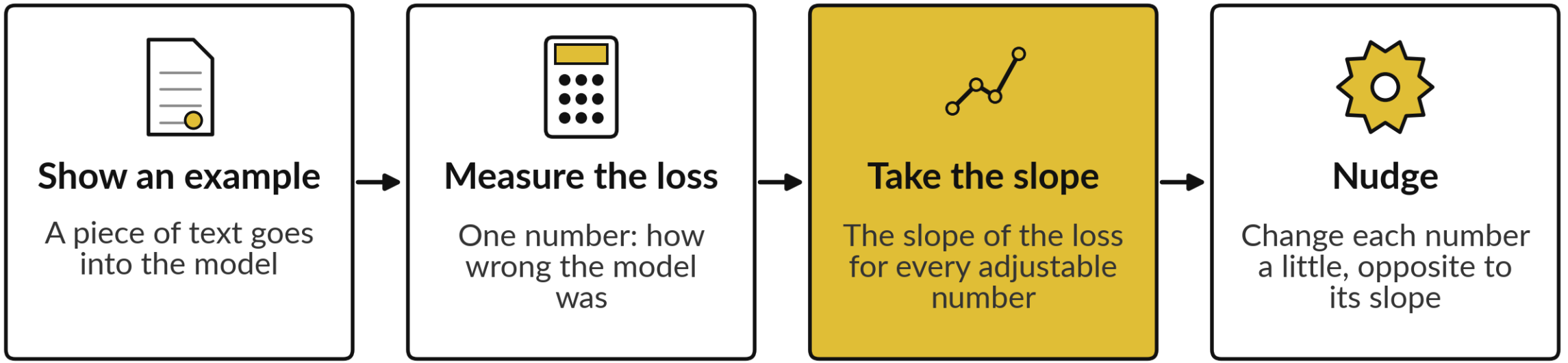
What you compute	You should see
Slope of x squared at 3, forward difference with h = 1	7
Same, with h = 0.01	6.01
Slope of x cubed at 2, central difference	12
Slope of x squared times e to the x at 1	8.155
Slope of x squared / (x + 1) at 1	0.75
Loss L(1), L(2)	11.75, 0.75
Slope of the loss at w = 1 and at w = 3	-18.5 and 11.5
Best w (slope zero)	2.233

If your number differs, stop and ask. Do not change a number to match. Find the step that went wrong.

Errors you will meet, what they mean and what to do

	What you see	What it means	What to do
	**NameError in Python**	A cell that defines a name was skipped	Run all cells from the top, in order
	**ZeroDivisionError or #DIV/0!**	The step h is 0 or the h cell is empty	Set h to a small number such as 0.001
	**Slope is 0.0 or full of noise**	h is far too small, for example 1e-12	Use h of about 0.001 to 0.00001
	**Slope has the wrong sign**	You subtracted in the wrong order	Use f(x + h) minus f(x), never the reverse

Where this appears in AI: the slope of the loss decides how each number changes



Published research describes language models of this kind as trained by this loop, repeated a very large number of times. The exact training details of ChatGPT, Gemini and Claude are not published.

Check an AI assistant's answer with your own numbers

8 minutes

Hands-on, with any assistant

Ask an assistant for the derivative of x squared times e to the x, and its value at x = 1.

Write down its answer and its value at x = 1.

Compare with ours: 3e = 8.155 at x = 1.

Run your lab slope function at x = 1.

If it differs, find the first line that differs.

You keep

One line: its value, your value, match or not. Never paste confidential data into an assistant.

Exit check: three questions with short answers

Question	Short answer
The slope of the loss at w is -4. Which way do you move w?	Up, so raise w. A negative slope means the loss falls as w grows.
What is the derivative of 5 x cubed at x = 2?	15 x squared, which is 60 at x = 2.
For f(x) = x cubed, a finite difference gives 12.0003 at x = 2 and the rule gives 12. Pass or fail?	Pass. They agree to 3 decimals. A finite difference is only an estimate.

Write your own answer before you read ours. The automatic check in your notebook or workbook counts the lab numbers.

Save the lab and keep three slope numbers

What you keep from today

Keep your saved lab. Write down three numbers: the loss slope -18.5 at w = 1, +11.5 at w = 3, and the best w = 2.233.

Our question today

A model has a number it can adjust. How does it know whether to raise it or lower it? (Session 10 answers how much.)

Next session

Session 8: partial derivatives and the gradient.

Take away

Saved lab
Notebook or workbook

Three numbers
-18.5, +11.5, 2.233

Next session
The gradient

Bring
Your slope function